

ShivaAI Jarvis Ultimate Architecture Specification

Autonomous AI Operating System (v5+ / v6 Vision)

1. Vision

ShivaAI Jarvis is a modular, enterprise-grade AI Operating System designed to function as:

- Personal AI Assistant
- Enterprise AI Copilot
- Salesforce Copilot
- Software Development Copilot
- Business Analyst Copilot
- Trading Intelligence Platform
- Research Assistant
- Knowledge Management System
- Workflow Automation Platform
- Digital Twin
- Real-World Intelligence Platform
- Future Humanoid Cognitive System

Core Principle:

One AI
Unlimited Capabilities
Fully Modular
Privacy First
Human Controlled

2. Core Design Principles

P1. Modular First

Every feature is a module.

P2. Privacy First

Sensitive features disabled by default.

P3. Human In Control

AI never exceeds granted permissions.

P4. Explainability

Major decisions must be explainable.

P5. Goal Driven

Focus on goals, not prompts.

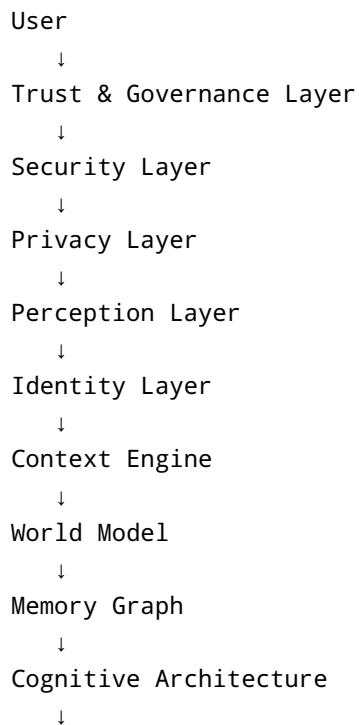
P6. Memory Enabled

Persistent learning and context.

P7. Enterprise Ready

Security, compliance, governance.

3. High-Level Architecture



```
Capability Registry
↓
Skill Graph
↓
Decision Intelligence
↓
Workflow Engine
↓
Execution Engine
↓
Modules
```

4. Modular Architecture Framework

Core Rule

Everything must be modular.

Module Categories

- Voice
- Vision
- Memory
- Planning
- Trading
- Salesforce
- Development
- Business Analysis
- Workflow
- Search
- Robotics
- Analytics
- Enterprise
- Emotional Intelligence

Feature Flags

```
voice: true
vision: false
memory: true
emotion_ai: false
robotics: false
```

```
trading: true  
salesforce: true
```

Dynamic Module Loading

Modules load only when required.

Benefits:

- Lower costs
 - Better security
 - Better performance
-

5. Cognitive Architecture Layer

Components

Intent Engine

Detects user intent.

Goal Engine

Converts requests into objectives.

Planning Engine

Creates execution plans.

Decision Engine

Evaluates options.

Reflection Engine

Reviews outputs.

Learning Engine

Continuously improves.

6. Unified Memory Graph

Memory Types

Short-Term Memory

Current conversation.

Session Memory

Current task.

Long-Term Memory

Historical interactions.

Knowledge Graph Memory

Relationship-based storage.

Stored Data

- Projects
 - Stories
 - Documents
 - Meetings
 - Decisions
 - Workflows
 - Preferences
-

7. Dynamic Capability Registry

Purpose:

Dynamically activate capabilities.

Registries

Domain Registry

- Development

- Trading
- Enterprise
- Research
- Business Analysis

Skill Registry

- Story Generation
- Coding
- Forecasting
- Planning

Tool Registry

- GitHub
- Jira
- Salesforce
- Slack

Workflow Registry

Reusable workflows.

8. Skill Graph Engine

Example:

```
Document Parsing
  ↓
Requirement Extraction
  ↓
Story Generation
  ↓
UI Generation
  ↓
Code Generation
  ↓
Deployment
```

9. Universal Context Engine

Combines:

- Chat History
 - Memory
 - Files
 - Meetings
 - Calendar
 - Emails
 - Knowledge Base
 - Enterprise Systems
-

10. Emotional Intelligence Engine

Purpose:

Understand emotions, communication style, urgency, and human context.

Features

Emotion Detection

Optional and consent-based.

Detect:

- Happy
 - Neutral
 - Frustrated
 - Concerned
 - Excited
 - Urgent
-

Communication Adaptation

Adjust:

- Tone
- Detail level
- Response style

Burnout Detection

Signals:

- Meeting overload
- Long work hours
- Repeated stress indicators

Team Sentiment Analysis

Analyze:

- Meetings
- Feedback
- Project communication

Emotional Memory

Store user preferences without storing sensitive psychological profiles.

11. Goal-Based Operating System

Convert goals into execution plans.

Example:

Launch ShivaAI v6

Creates:

- Roadmap
 - Stories
 - Tasks
 - Risks
 - Milestones
-

12. Autonomous Workflow Engine

Supports:

- Triggers
- Conditions
- Actions
- Scheduling

Example:

```
BRD Uploaded
↓
Extract Requirements
↓
Generate Stories
↓
Create Jira Tasks
```

13. Decision Intelligence Layer

Every major action receives:

- Confidence Score
- Risk Score
- Cost Score
- Impact Score

14. Reflection & Self-Correction

Process:

```
Generate
↓
Review
↓
Improve
↓
Deliver
```

15. Execution Verification Layer

Checks:

- Success
 - Failure
 - Logs
 - Health Status
-

16. Real-World Intelligence Framework

Purpose:

Enable perception and environment understanding.

Vision Intelligence

Capabilities:

- OCR
 - Object Detection
 - Scene Understanding
 - Video Understanding
-

Voice Intelligence

Capabilities:

- Speech Recognition
 - Speaker Recognition
 - Voice Commands
-

Activity Recognition

Detect:

- Coding
- Meeting

- Trading
 - Learning
 - Research
-

Environmental Awareness

Understand:

- Home
 - Office
 - Meeting Rooms
 - Devices
-

17. Identity Intelligence Layer

Purpose:

Understand authorized users.

Methods:

- Login Session
- Device Identity
- Voice Recognition (Optional)
- Face Recognition (Optional)

All biometric features must be opt-in.

18. World Model Engine

Maintain digital representation of:

- Users
 - Teams
 - Projects
 - Devices
 - Locations
 - Systems
-

19. Human Digital Twin

Learns:

- Preferences
- Habits
- Working Style
- Goals

Uses:

- Recommendations
 - Planning
 - Automation
-

20. Search Intelligence Layer

Unified search across:

- Memory
 - Files
 - Email
 - Jira
 - Salesforce
 - Web
-

21. Enterprise Knowledge Hub

Integrations:

- Salesforce
 - Jira
 - Confluence
 - ServiceNow
 - SharePoint
 - Slack
 - Teams
-

22. Salesforce Copilot

Capabilities:

- Apex Generation
 - LWC Generation
 - Flow Creation
 - Object Design
 - SOQL Optimization
 - Deployment Assistance
-

23. Development Copilot

Capabilities:

- Architecture Design
 - Code Generation
 - Testing
 - Documentation
 - CI/CD Support
 - DevOps Assistance
-

24. Business Analyst Copilot

Capabilities:

- BRD Analysis
 - Story Generation
 - Acceptance Criteria
 - Process Mapping
 - Wireframe Suggestions
-

25. Trading Intelligence Platform

Modules:

- Stocks
- Forex
- Crypto
- Futures

- Options

Features:

- Portfolio Analysis
 - Risk Engine
 - Sentiment Analysis
 - Strategy Testing
 - Simulation
-

26. Meeting Intelligence

Features:

- Transcription
 - Summaries
 - Action Items
 - Jira Story Creation
 - Follow-up Emails
-

27. AI App Builder

Generate:

- Frontend
 - Backend
 - Database
 - APIs
 - Tests
 - Deployment Scripts
-

28. Robotics & Humanoid Readiness

Future Layer

Capabilities:

- Navigation
- Manipulation
- Object Handling
- Environmental Awareness

Technologies:

- ROS2
 - Isaac ROS
 - SLAM
-

29. Trust, Privacy & Governance

AI Constitution

Rules:

- Protect Privacy
 - Respect Permissions
 - Explain Decisions
 - Never Expose Secrets
-

Privacy Controls

Users can disable:

- Camera
 - Microphone
 - Voice Identification
 - Face Recognition
 - Activity Tracking
-

Data Isolation

Separate:

- Personal Data
 - Enterprise Data
 - Public Data
-

Consent Management

Required for:

- Biometrics

- Recording
 - Monitoring
-

30. Security Framework

Zero Trust Architecture

Everything validated.

RBAC

Roles:

- Admin
 - Manager
 - Developer
 - Analyst
 - Viewer
-

ABAC

Access based on:

- User
 - Team
 - Project
 - Location
-

Secret Management

- Vault
 - AWS Secrets Manager
 - Azure Key Vault
-

Encryption

AES-256

31. Audit & Explainability

Every action logged.

Every decision explainable.

32. Deployment Architecture

Personal

Voice + Memory + Productivity

Developer

Code + GitHub + Documentation

Enterprise

Salesforce + Jira + Knowledge Hub

Trading

Market Data + Analytics

Robotics

Vision + Voice + Navigation

33. Developer Roadmap

Phase 1

- Cognitive Architecture
- Memory Graph
- Capability Registry
- Workflow Engine

- Voice Assistant
-

Phase 2

- Vision
 - Search
 - Meeting Intelligence
 - Enterprise Integrations
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Phase 3

- Digital Twin
 - World Model
 - Emotional Intelligence
 - Decision Intelligence
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Phase 4

- Smart Home
 - IoT
 - Ambient Intelligence
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Phase 5

- Robotics
 - Humanoid Integration
 - Autonomous Physical Actions
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34. Success Metrics

Track:

- Task Success Rate
- Workflow Success Rate
- User Satisfaction
- Automation Coverage
- Response Accuracy
- Cost Per Task
- Security Incidents

- Time Saved
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Final Vision

ShivaAI Jarvis is not a chatbot.

It is a Modular AI Operating System that can perceive, reason, remember, plan, learn, collaborate, automate, and eventually interact with the physical world while remaining secure, privacy-preserving, explainable, and fully controlled by the user.